

第五讲：特征描述与图像对齐



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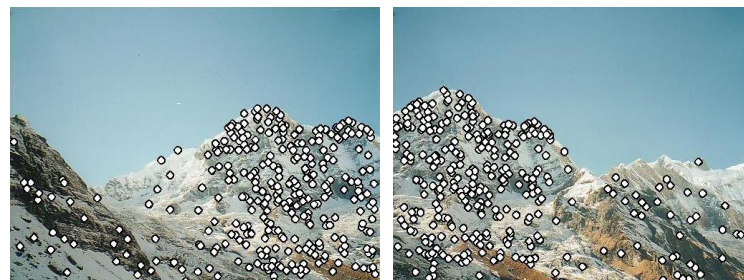
电子科技大学机械与电气工程学院

2026-3-17

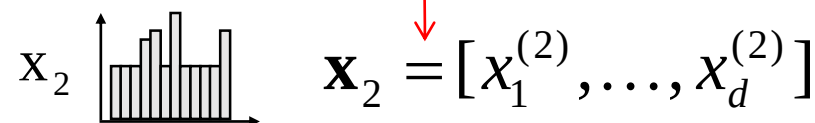
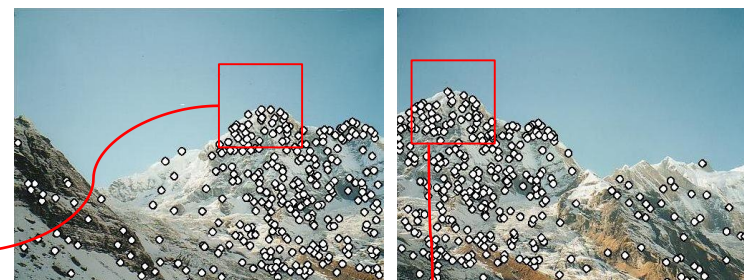
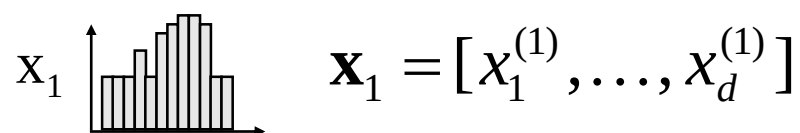


Local features: main components

- 1) Detection(特征检测):
Find a set of distinctive key points.



- 2) Description (特征描述):
Extract feature descriptor around each interest point as vector.



- 3) Matching (特征匹配):
Compute distance between feature vectors to find correspondence.

$$d(\mathbf{x}_1, \mathbf{x}_2) < T$$



Outline



1, Local Image Descriptors

2, Image alignment

Basic 2D Transformations

Transform parameter fitting

RANSAC

Image Representations: Histograms



Global histogram to represent distribution of features

– Color, texture, depth, ...

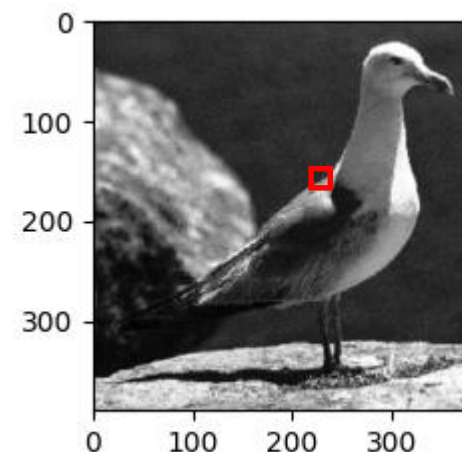
Local histogram per detected point

基于局部直方图的特征描述

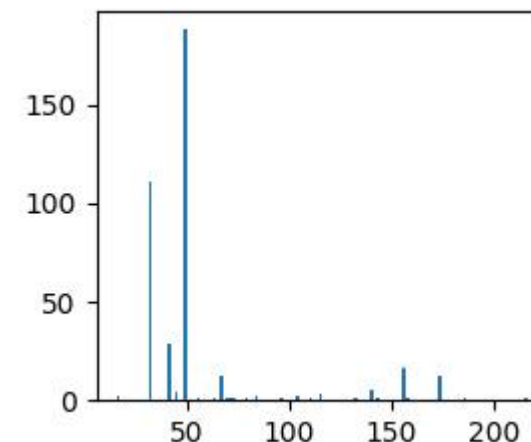
Drawbacks:

– Sensitive to illuminance and orientation

对光照和方向敏感



直方图



Histogram of gradients 梯度直方图

- 基于局部梯度直方图的特征描述
- 尺度不变特征转换 (SIFT: Scale Invariant Feature Transform)
 - 广泛使用的特征描述方法
 - 具有方向不变性, 尺度不变性
 - 对光照变化不敏感

Object recognition from local scale-invariant features

DG Lowe - ... IEEE international conference on computer vision, 1999 - ieeexplore.ieee.org

... new method for image feature generation called the Scale Invariant Feature Transform (SIFT). This approach transforms an image into a large collection of local feature vectors, each of ...

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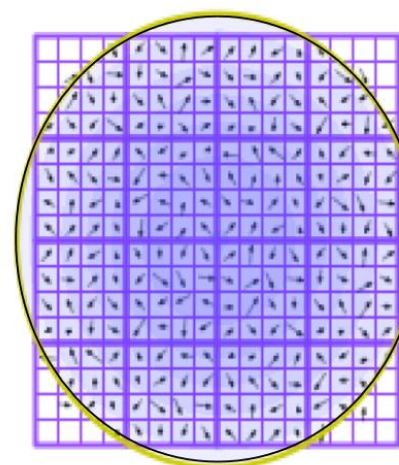
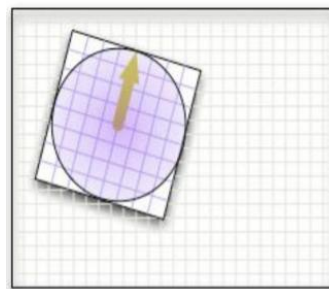
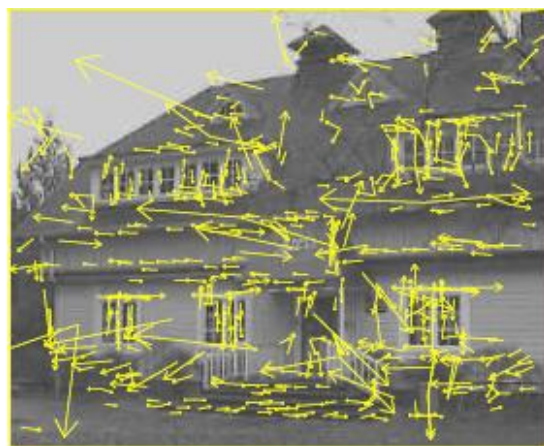
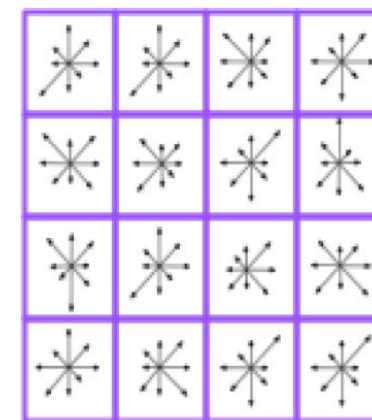


Image gradients



Keypoint descriptor

Scale invariant feature transform (SIFT)



尺度不变特征变换

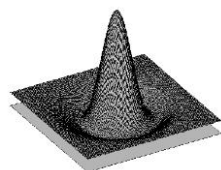
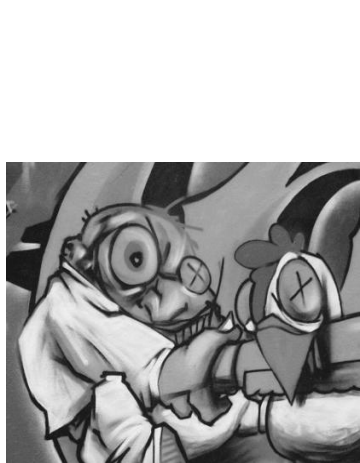
- Find local maxima in position-scale space

Object recognition from local scale-invariant features

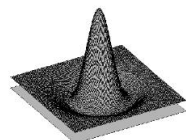
DG Lowe - ... IEEE international conference on computer vision, 1999 - ieeexplore.ieee.org

... new method for image feature generation called the Scale Invariant Feature Transform (SIFT). This approach transforms an image into a large collection of local feature vectors, each of ...

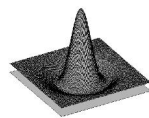
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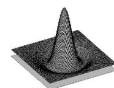
σ^5



σ^4



σ^3



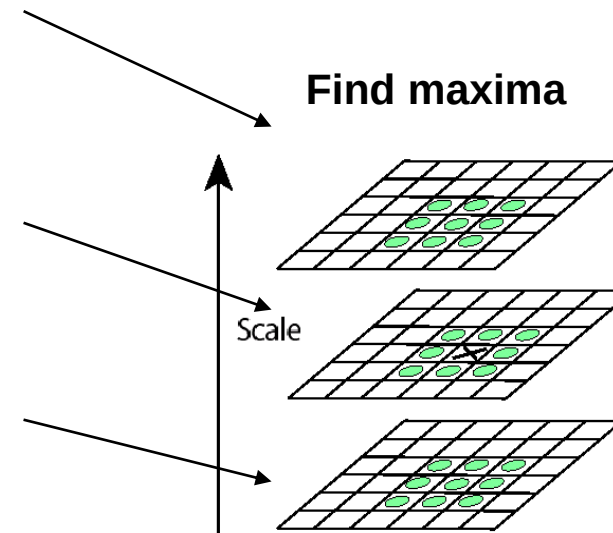
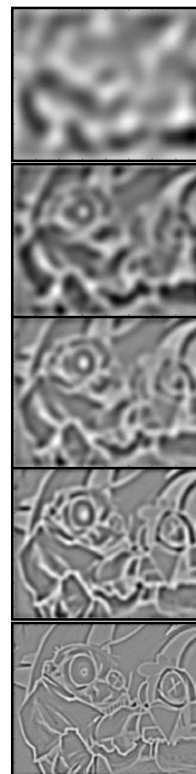
σ^2



σ

在多个尺度上进行滤波，
取不同尺度上最大的响应

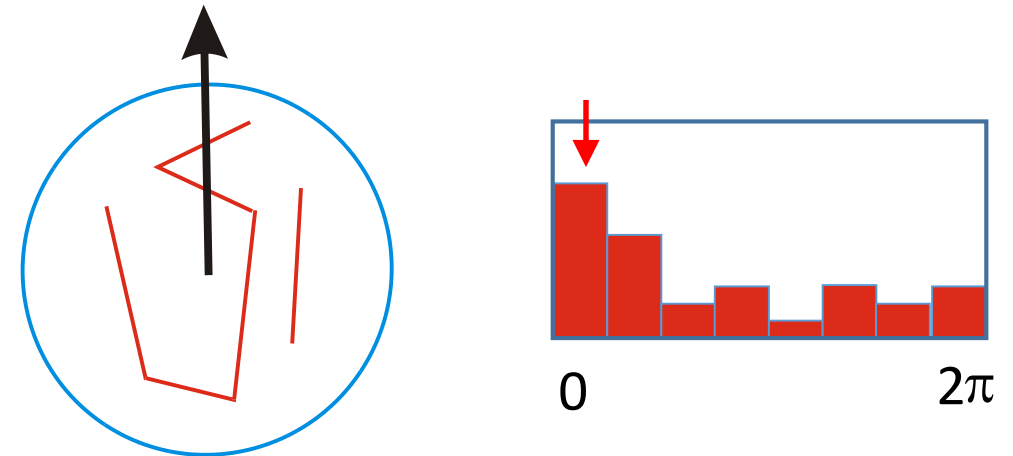
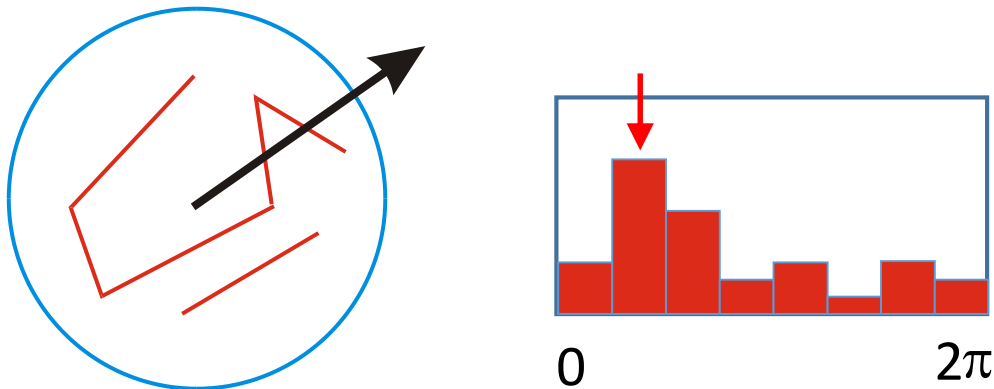
Difference of Gaussian (DoG)



⇒ List of
(x, y, s)

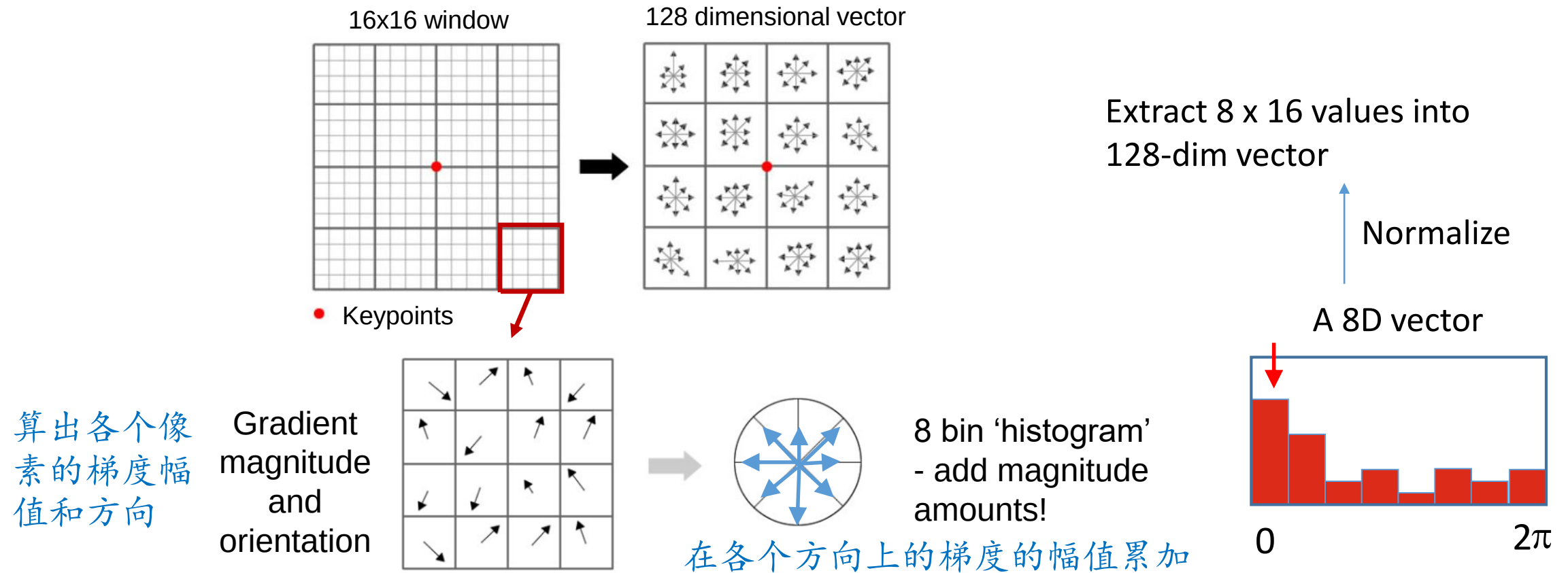
SIFT Orientation Estimation

- Compute gradient orientation histogram 计算梯度方向直方图
- Select dominant orientation θ 对齐到主导方向: 实现旋转不变性
- Normalize: rotate to fixed orientation



SIFT Descriptor Extraction

- Given a keypoint with scale and orientation

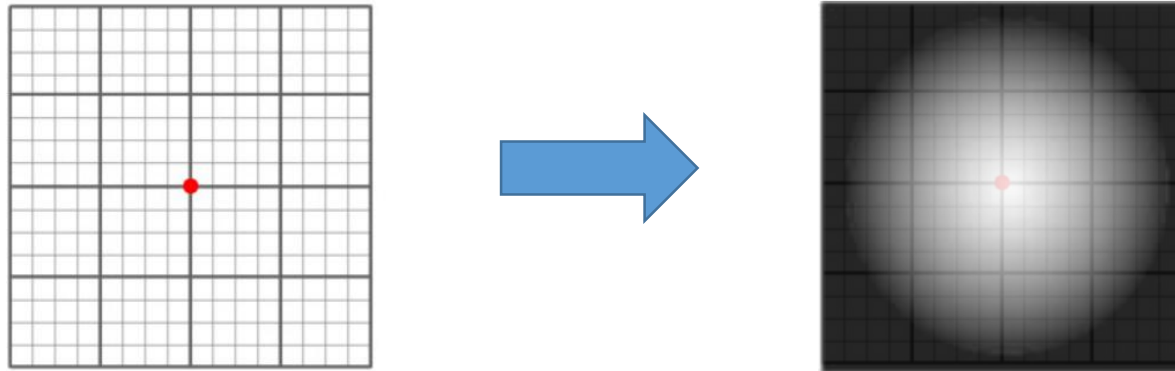


SIFT Descriptor Extraction

- Gaussian weight the magnitudes for smoothness

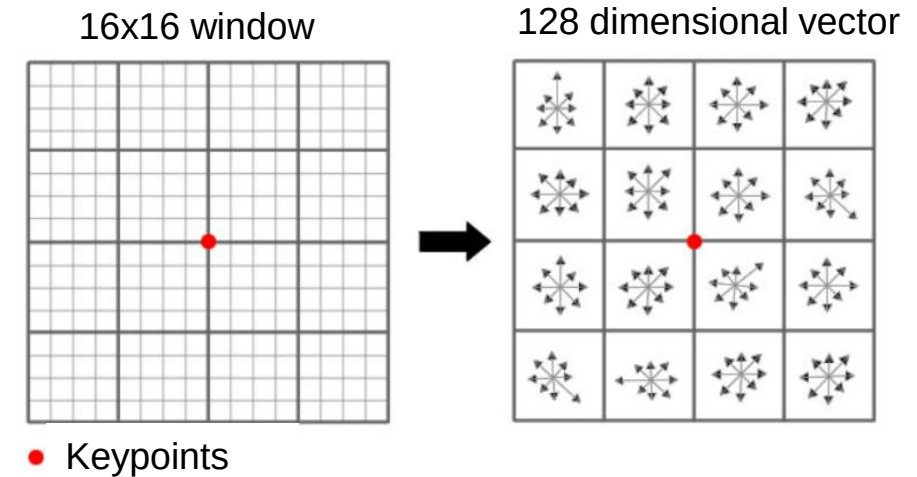
Weight 16x16 grid by Gaussian to add location robustness and reduce effect of outer regions

通过高斯函数对各个像素的梯度进行加权



SIFT Descriptor Extraction

- Extract 8 x 16 values into 128-dim vector
- Advantages:
 - **Illumination invariance:** 对光照变化不敏感
 - Working in gradient space, so robust to $I = I + b$
 - Normalize vector to $[0...1]$
 - Robust to $I = \alpha I$ brightness changes
 - **Rotation invariant** 具有旋转不变性



GLOH: Gradient location-orientation histogram

梯度位置朝向直方图

- **Log-polar bins** are used for orientation histograms
 - Instead of using square bins
- 272-dimensional histogram
 - Spatial bins based on radius of 6, 11, 15
 - 8 angular bins for each radius
 - 16 orientation bins

$$17 \times 16 = 272$$

A performance evaluation of local descriptors

[K Mikolajczyk](#), [C Schmid](#) - ... transactions on pattern analysis and ..., 2005 - [ieeexplore.ieee.org](#)
... of the art on **local descriptors**. Section 3 describes the implementation details for the detectors and **descriptors** used in our comparison as well as our **evaluation** criterion and the data set...

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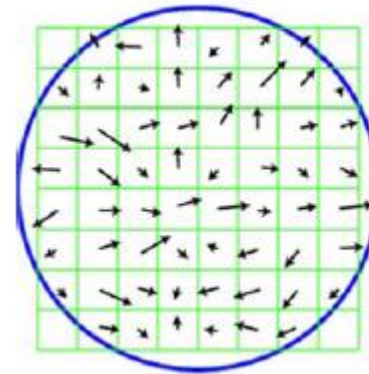
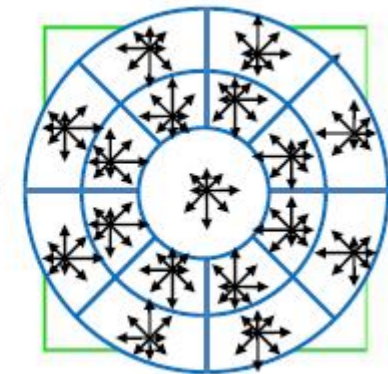


Image gradients

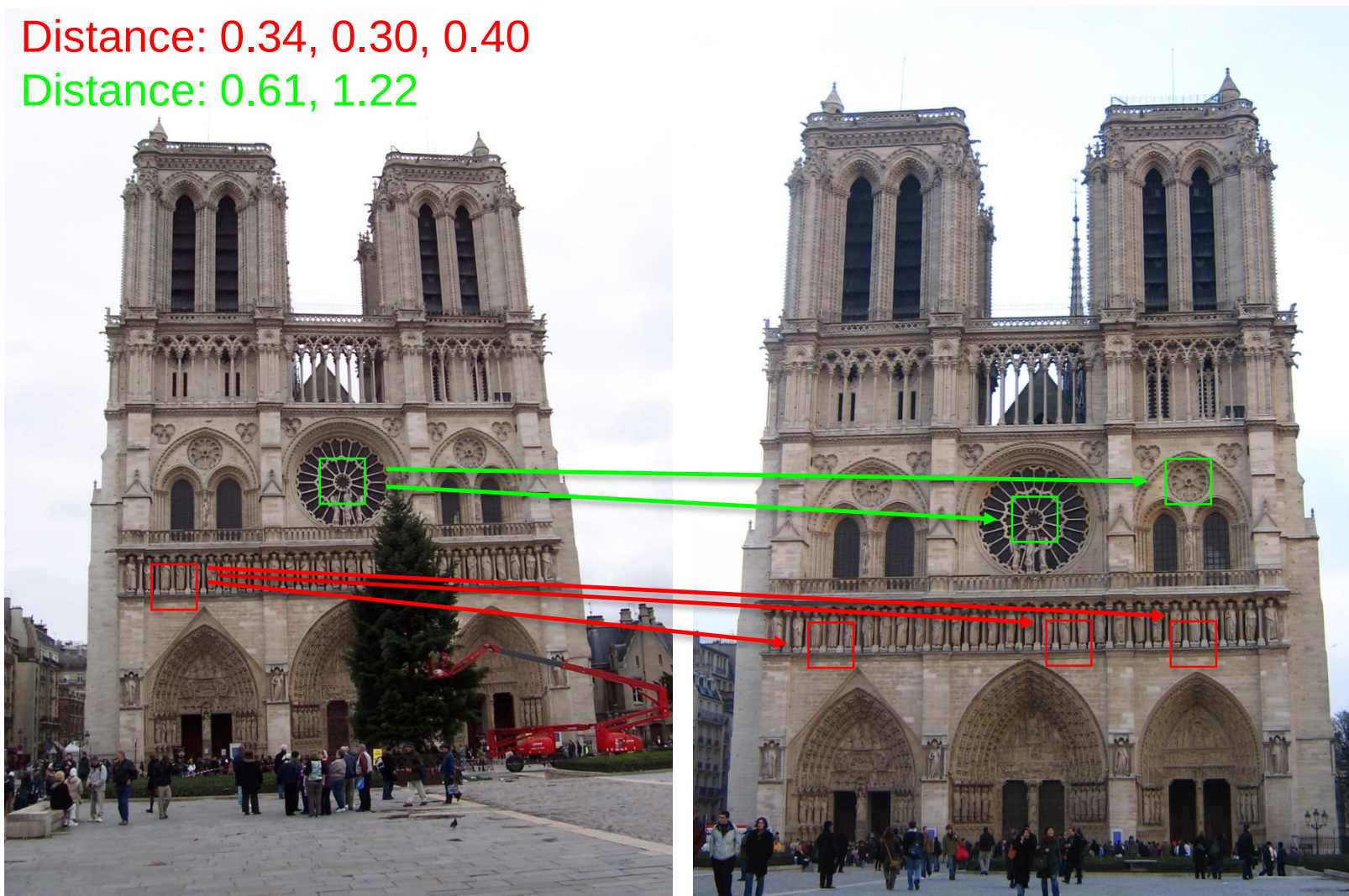


Keypoint descriptor

特征匹配：从两幅图像中找出最相似的特征点

Distance: 0.34, 0.30, 0.40

Distance: 0.61, 1.22



计算特征相似性：

- Euclidean distance
欧氏距离
- Cosine similarity
余弦相似度

Outline



1, Local Image Descriptors

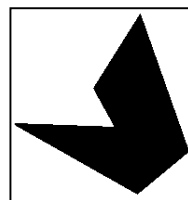
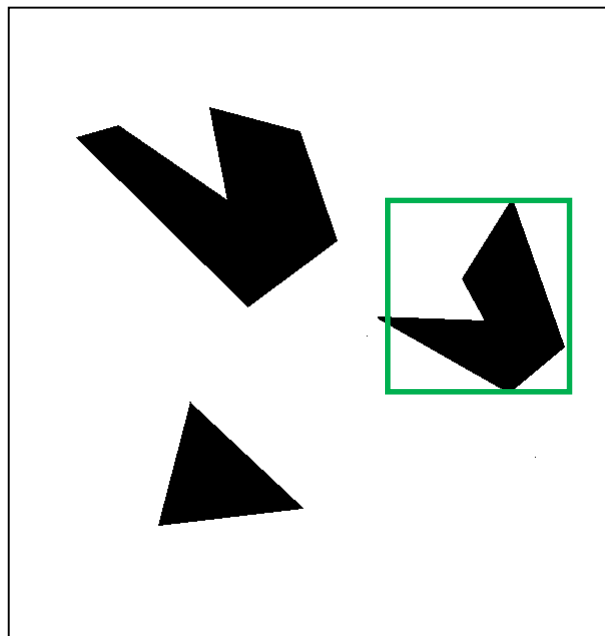
2, Image alignment

Basic 2D Transformations

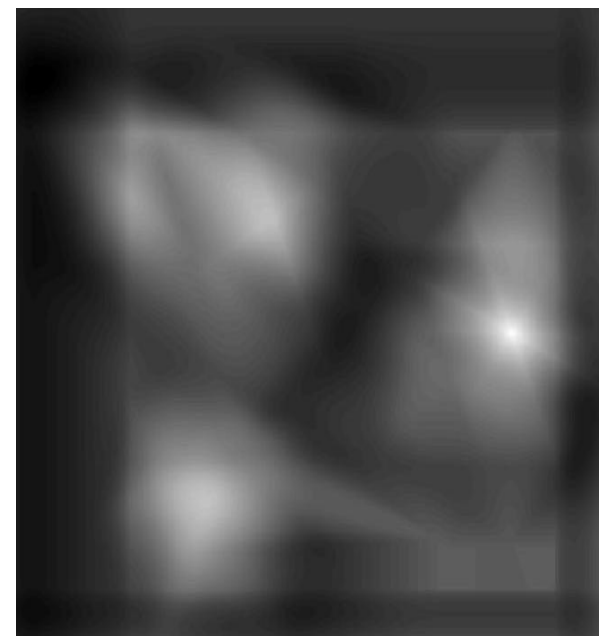
Transform parameter fitting

RANSAC

回顾：模板匹配



Template (mask)



Normalized cross-correlation (NCC)

要求：图像中的目标域模板的大小、方向和颜色值一致

Image alignment 图像对齐

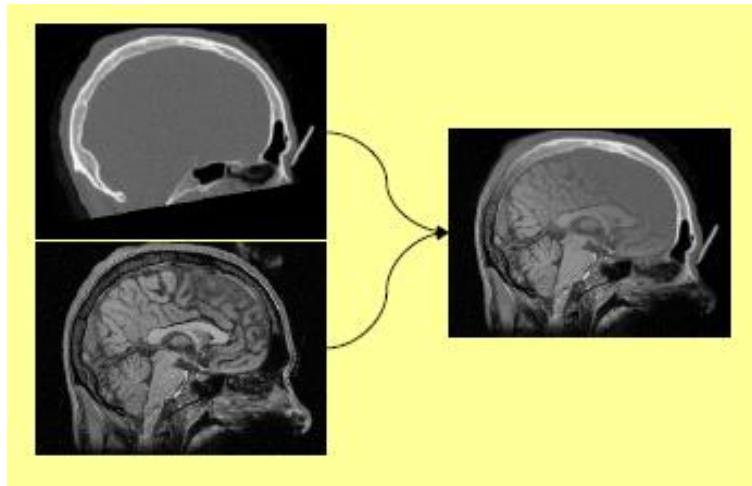
- Fitting an affine transformation
 - Affine model approximates perspective projection of planar objects.

- If the template and target are with the same size and orientation, we can use **cross-correlation**
- Otherwise, we need to estimate the **transformation** between them

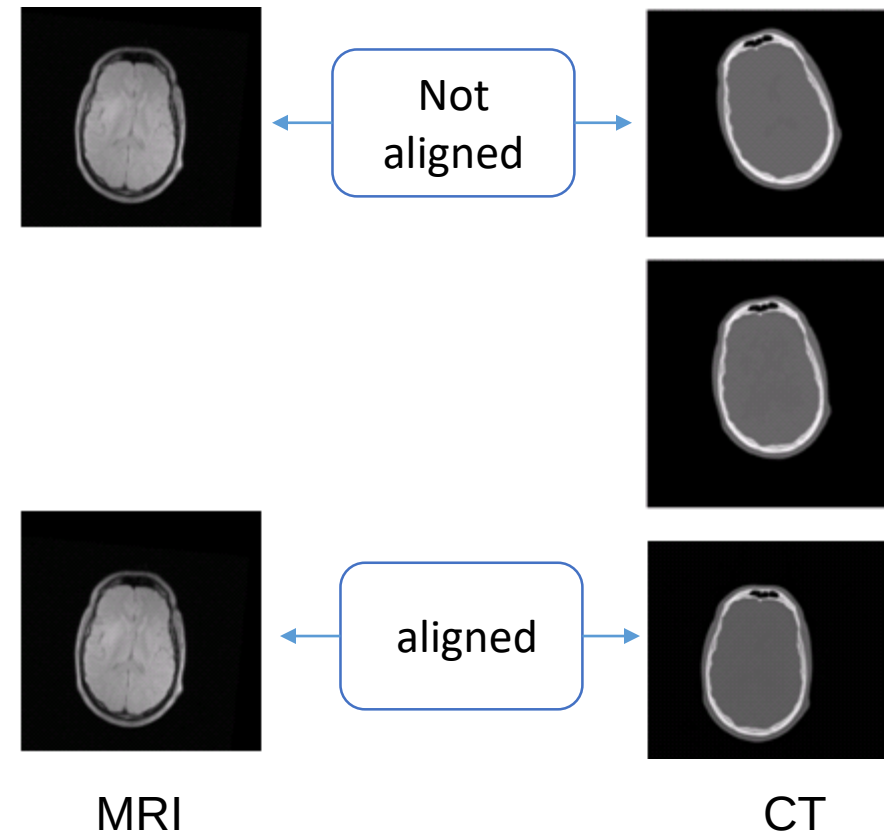


Alignment Examples

- Medical Image Registration 医学图像配准



CT has a good contrast for bones
MRI has a good contrast to soft tissues
We would like to align the CT and MRI of the same patient



Panoramas 全景图

- Obtain a wider angle view by combining multiple images.

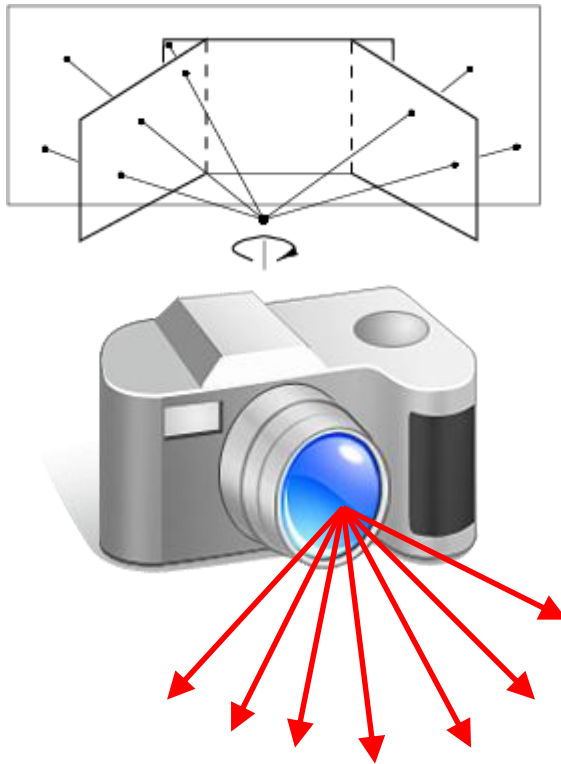
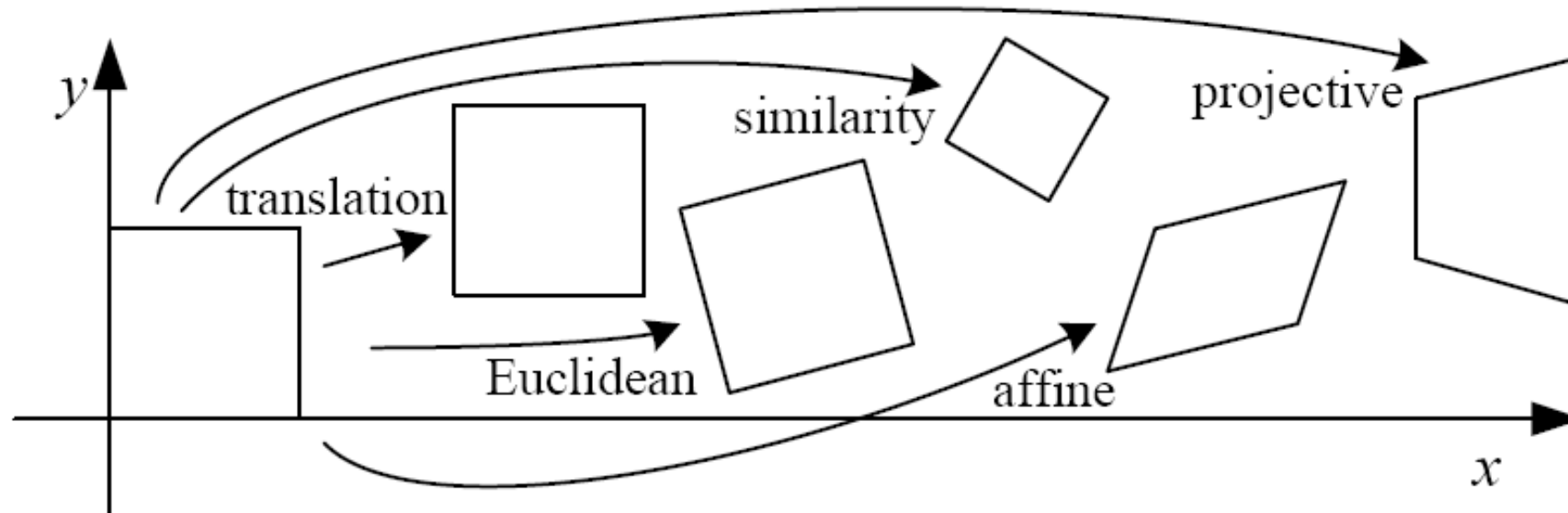


image from S. Seitz

Parametric (global) warping

- Basic 2D Transformations



Parametric (global) warping

- Examples of parametric warps



Original image



Translation 平移



Rotation 旋转



Aspect 拉伸



Affine
仿射变换



Perspective
投影变换



cylindrical

Basic 2D Transformations



- Basic 2D transformations as 3x3 matrices

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Scale

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Rotate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & sh_x & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Shear

Basic 2D Transformations

- 2D Affine Transformations
- Affine transformations are combinations of ...

- Linear transformations, and
- Translations

仿射变换矩阵（6个参数）

$$\begin{bmatrix} x' \\ y' \\ w \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

- Parallel lines remain parallel

仿射变换中，直线的平行关系可以保留



Original image



affine

Basic 2D Transformations



- 2D Projective Transformations

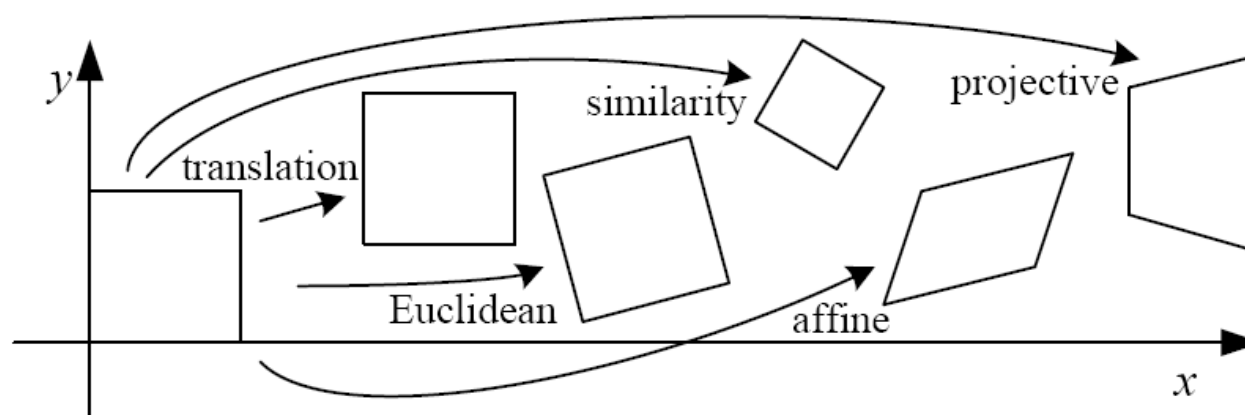
- Projective transformations:

- Affine transformations, and
- Projective warps

投影变换矩阵 (9个参数)

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

- Parallel lines do not necessarily remain parallel



投影变换中，直线的平行关系可能丢失

Image alignment 图像对齐

- Two broad approaches:
 - Direct (**pixel-based**) alignment 基于像素的对齐 费时，对颜色变化敏感
 - Search for alignment where most pixels agree
 - Feature-based** alignment 基于特征的对齐 更加高效，对颜色变化鲁棒
 - Search for alignment where extracted features agree
 - Can be verified using pixel-based alignment

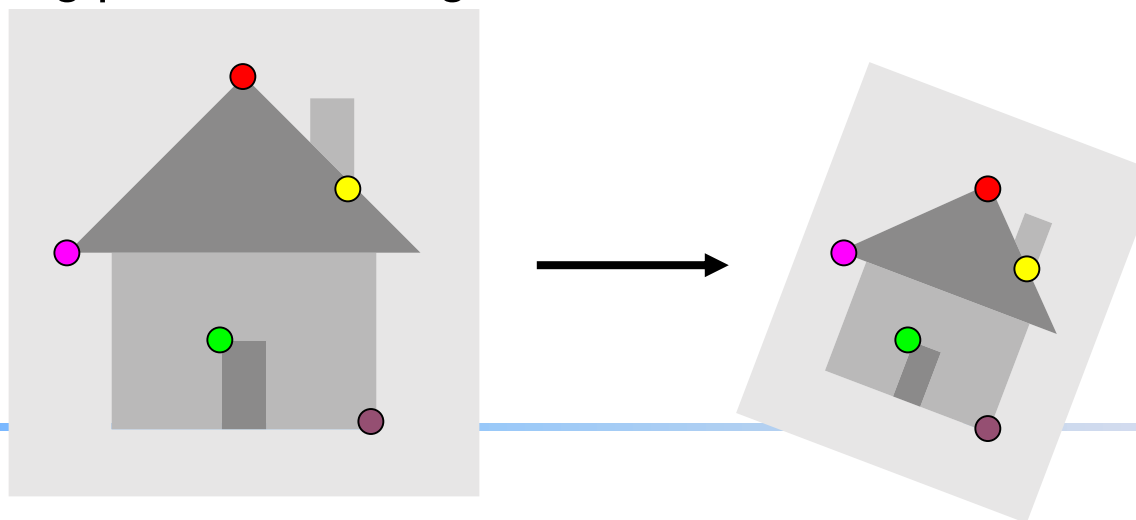


Image alignment 图像对齐

- Fitting an affine transformation
 - Assuming we know the correspondences, how do we get the transformation?

如何估计变换参数

$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

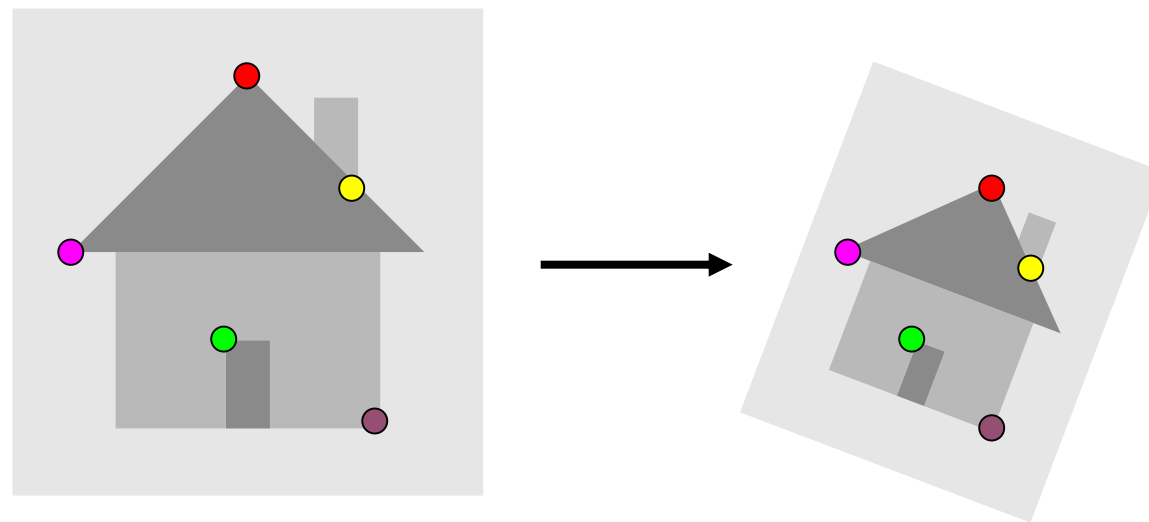


Image alignment



- Fitting an affine transformation
 - Assuming we know the correspondences, how do we get the transformation?

$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

$$\begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

Image alignment



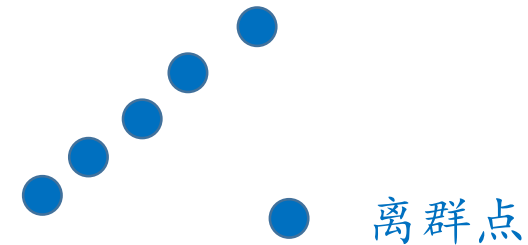
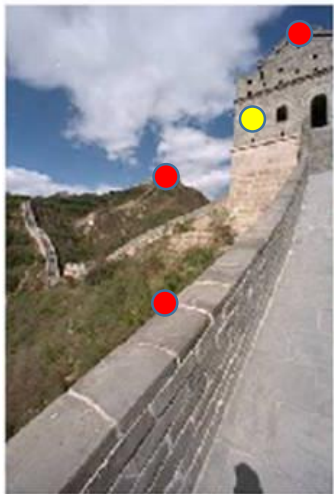
- Fitting an affine transformation
 - Assuming we know the correspondences, how do we get the transformation?

$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$
$$\begin{bmatrix} \dots & \dots & \dots & \dots & \dots & \dots \\ x_i & y_i & 0 & 0 & 1 & 0 \\ 0 & 0 & x_i & y_i & 0 & 1 \\ \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} \dots \\ x'_i \\ y'_i \\ \dots \end{bmatrix}$$

Can be solved by least square

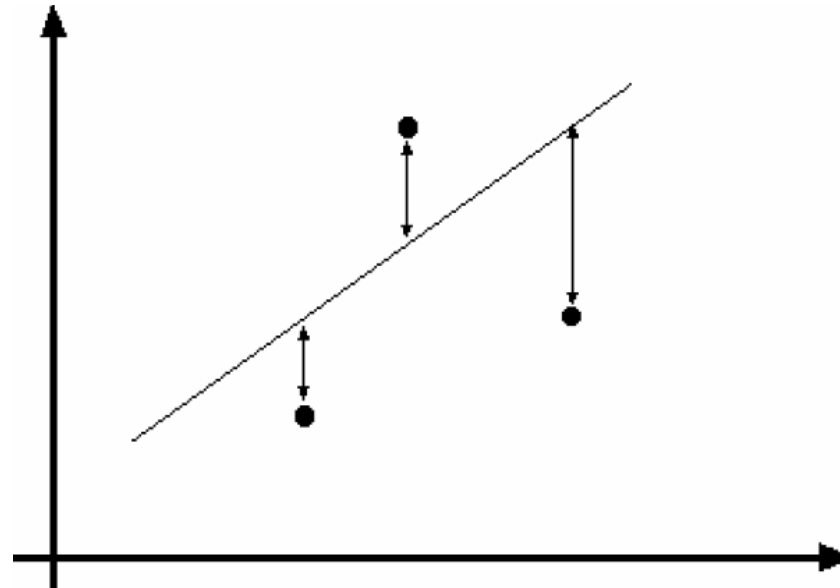
Transform parameter fitting

- **Outliers** can hurt the quality of our parameter estimates, e.g.,
 - an erroneous pair of matching points from two images
 - an edge point that is noise, or doesn't belong to the line we are fitting



Transform parameter fitting

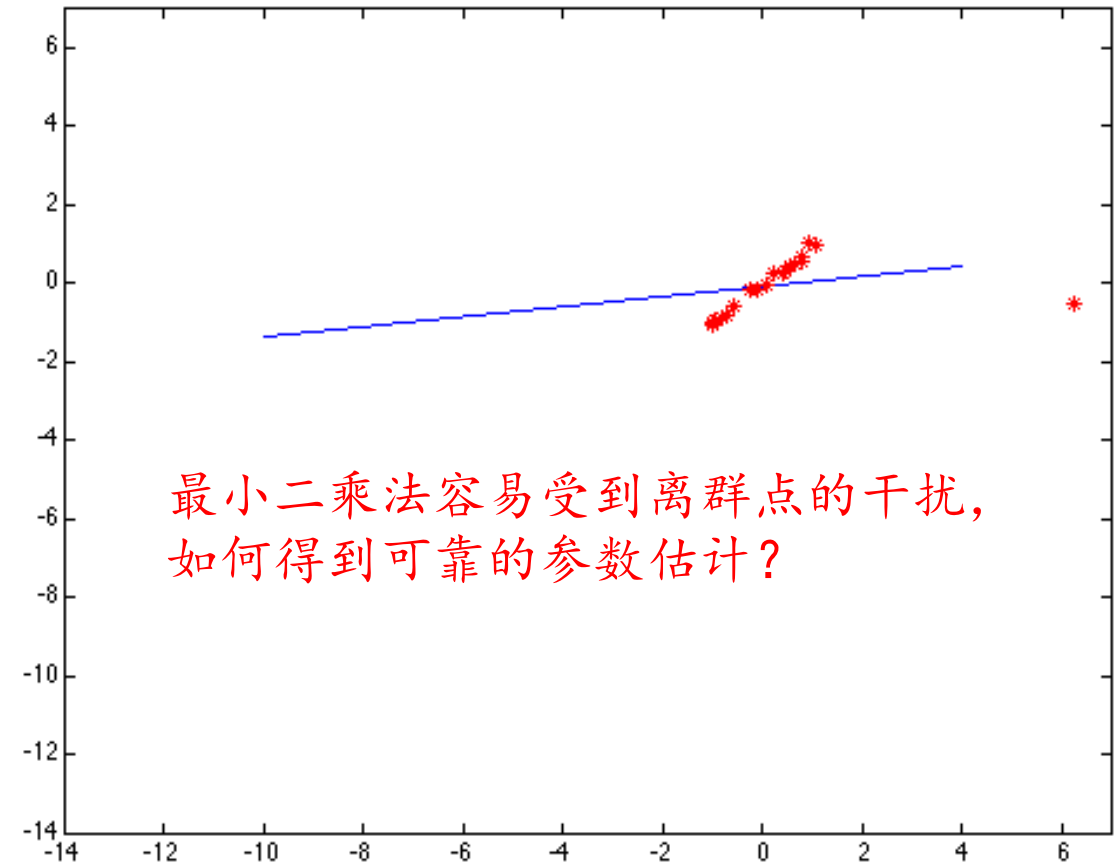
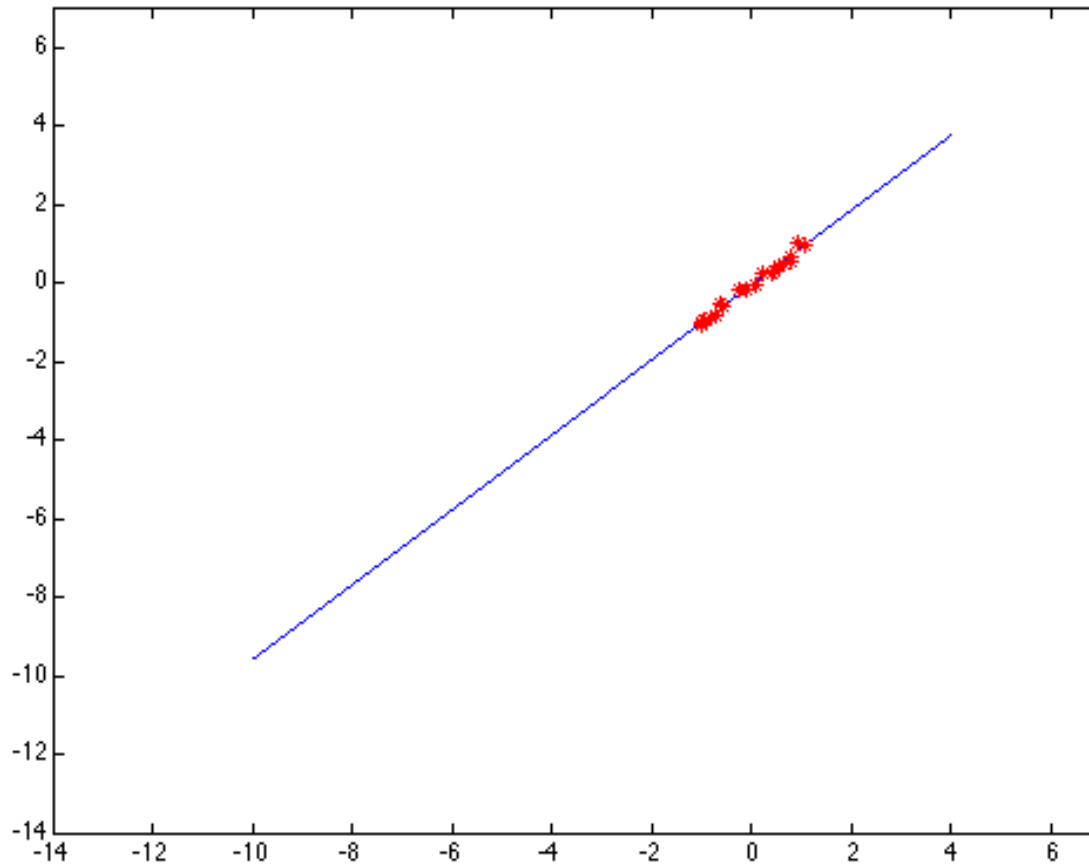
- Example: least squares line fitting
 - Assuming all the points that belong to a particular line are known



Transform parameter fitting



- Outliers affect least squares fit



Outline



1, Local Image Descriptors

2, Image alignment

Basic 2D Transformations

Transform parameter fitting

RANSAC 随机抽样一致性

Transform parameter fitting



- RANSAC: RANdom Sample Consensus 随机抽样一致性
- **Approach**: we want to avoid the impact of **outliers**, so let's look for “**inliers**”, and use those only.
- **Intuition**: if an outlier is chosen to compute the current fit, then the resulting line won't have much support from rest of the points.



Transform parameter fitting



- RANSAC: RANdom Sample Consensus

随机选择一组点

拟合变换参数

寻找与该变换相符的点

重新拟合参数

- 1) Randomly select a seed group of points on which to base transformation estimate (e.g., a group of matches)
- 2) Compute transformation from selected group
- 3) Find inliers to this transformation
- 4) If the number of inliers is sufficiently large, re-compute least-squares estimate of transformation on all of the inliers

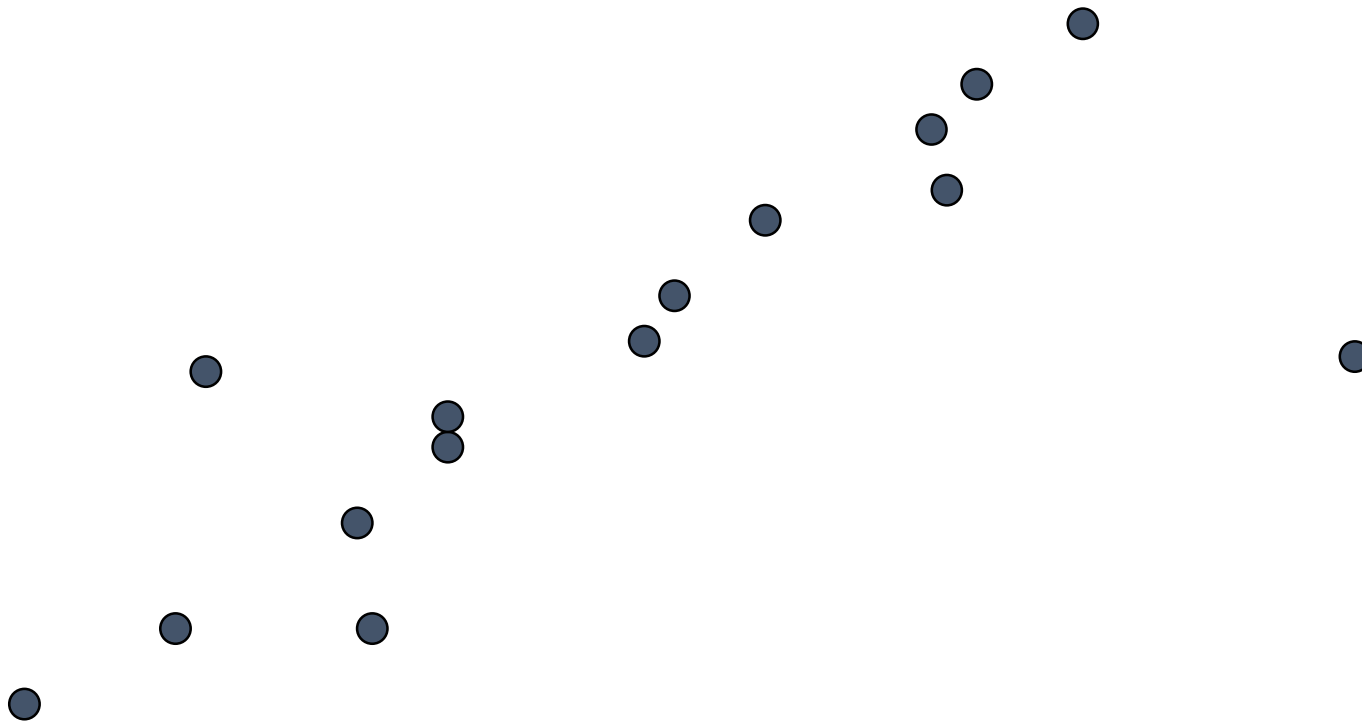
Keep the transformation with the largest number of inliers



Transform parameter fitting



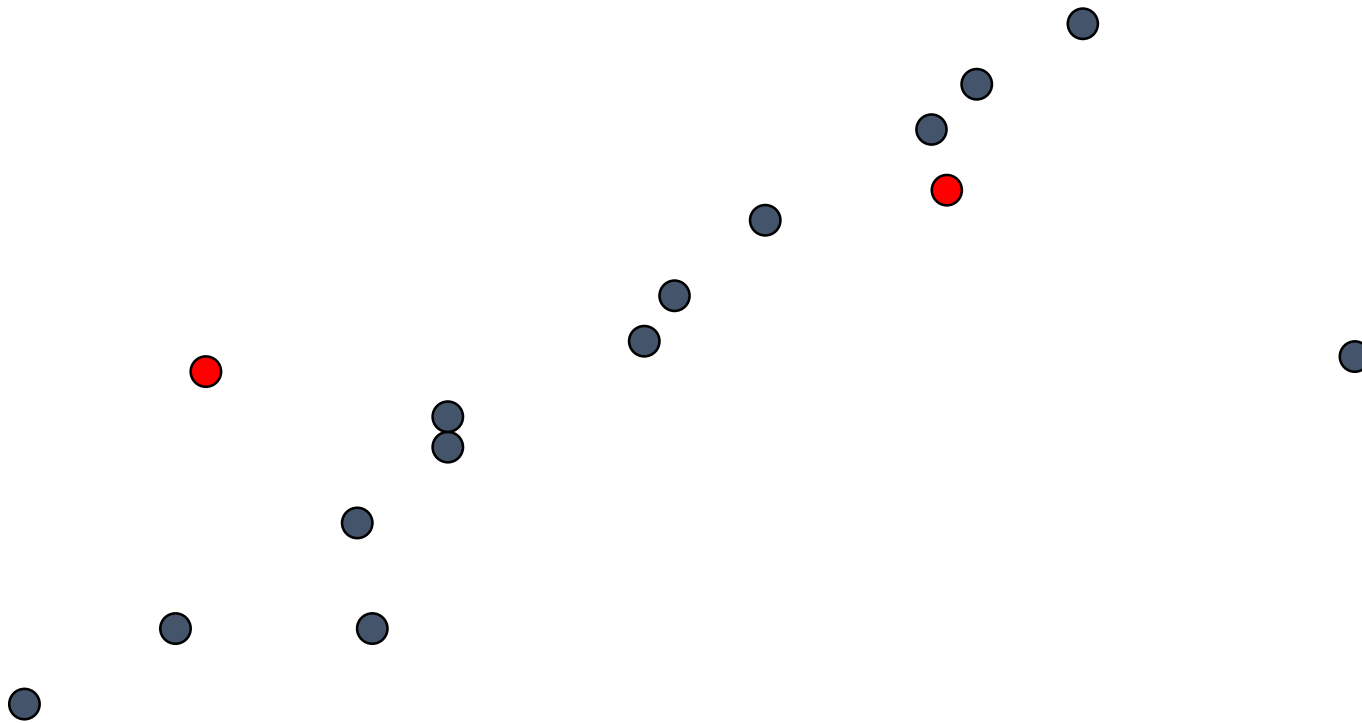
- RANSAC Line fitting example
- Task: estimate the best line



Transform parameter fitting

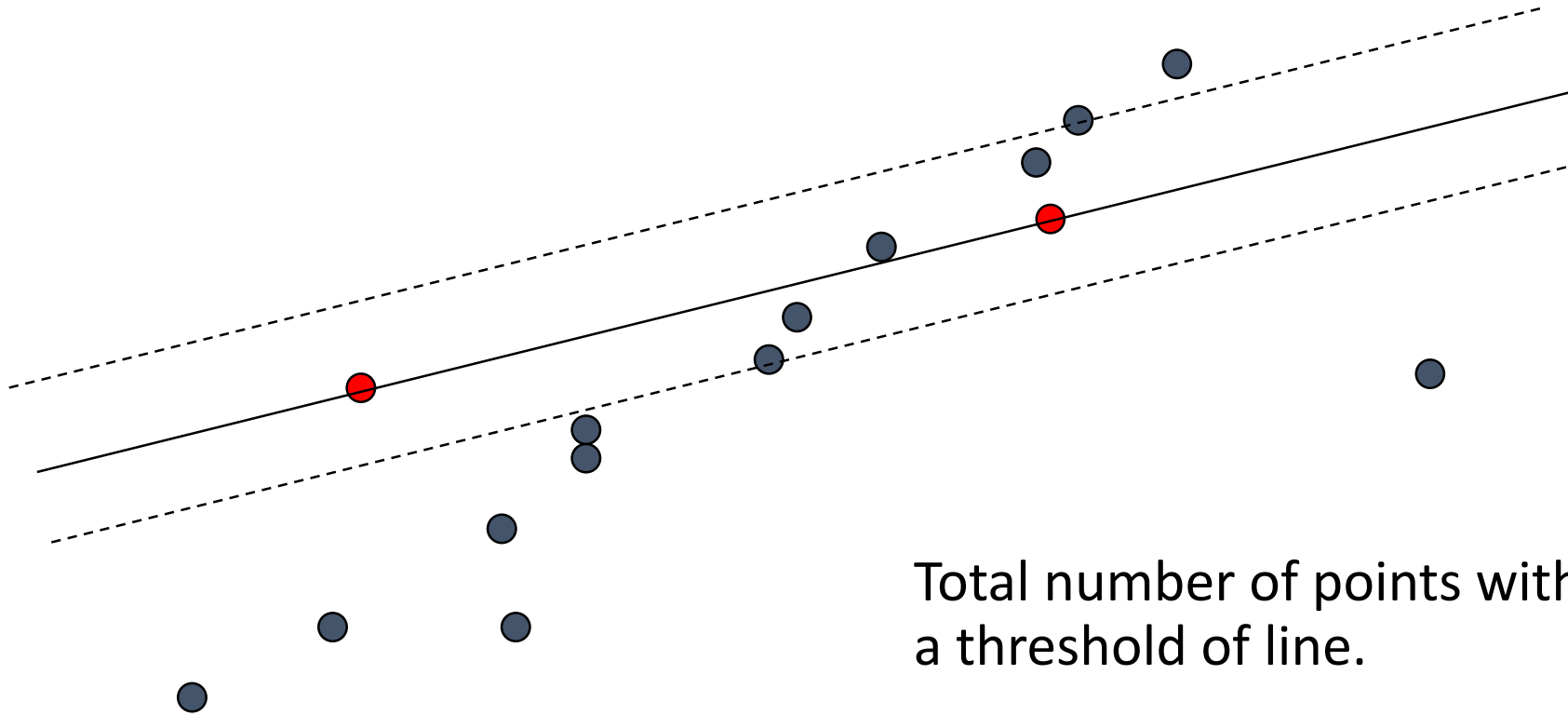


- RANSAC Line fitting example
- Sample two points



Transform parameter fitting

- RANSAC Line fitting example
- Find inliers

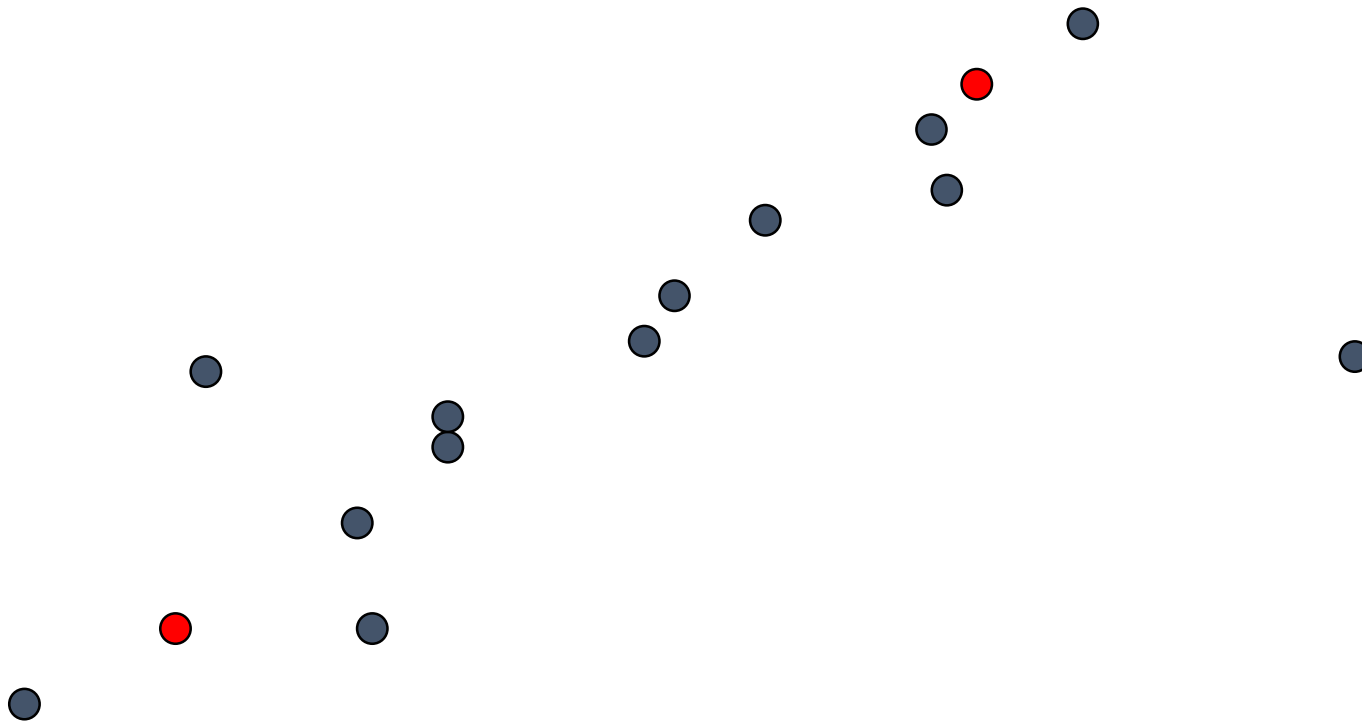


Total number of points within
a threshold of line.

Transform parameter fitting



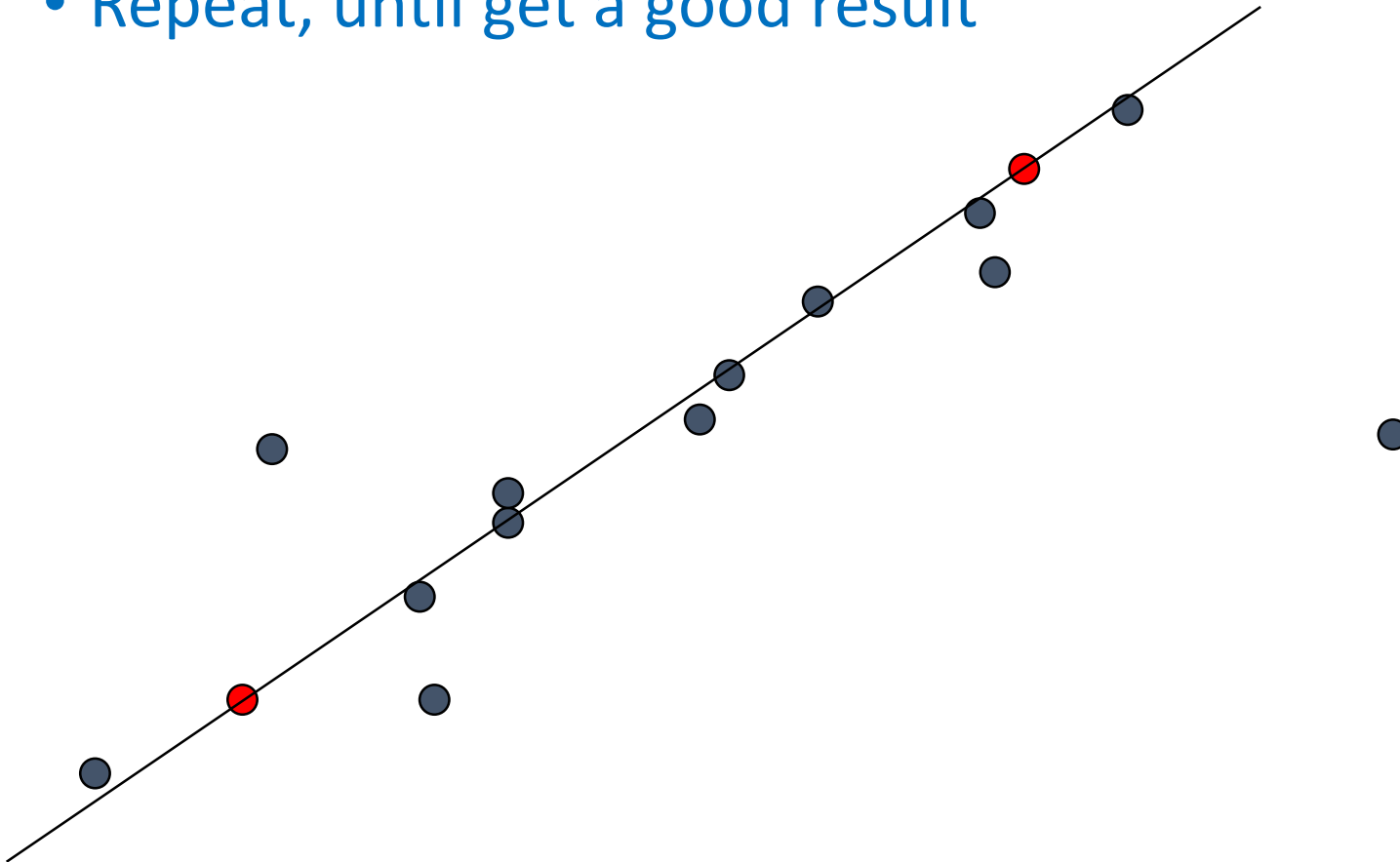
- RANSAC Line fitting example
- Repeat, until get a good result



Transform parameter fitting

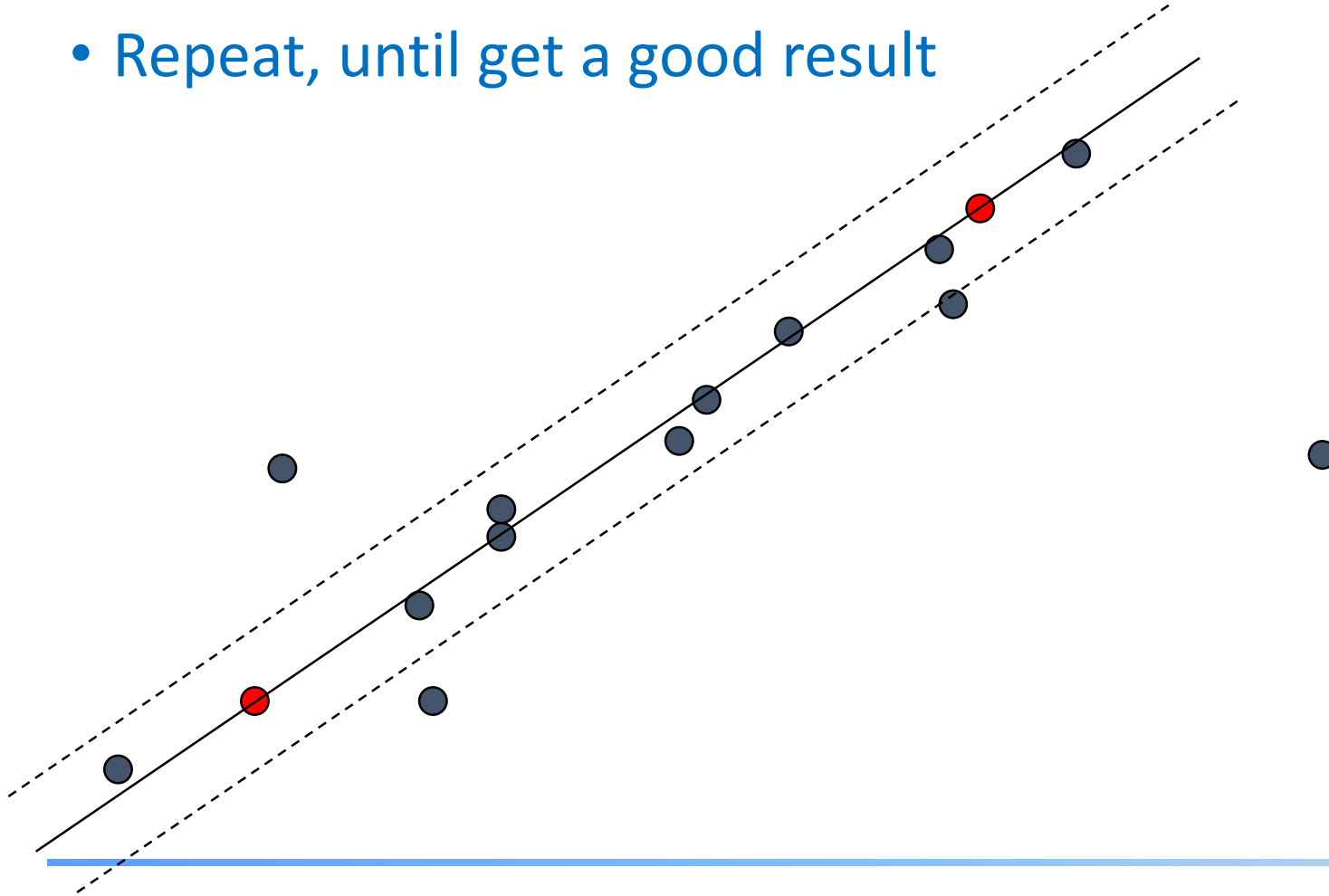


- RANSAC Line fitting example
- Repeat, until get a good result

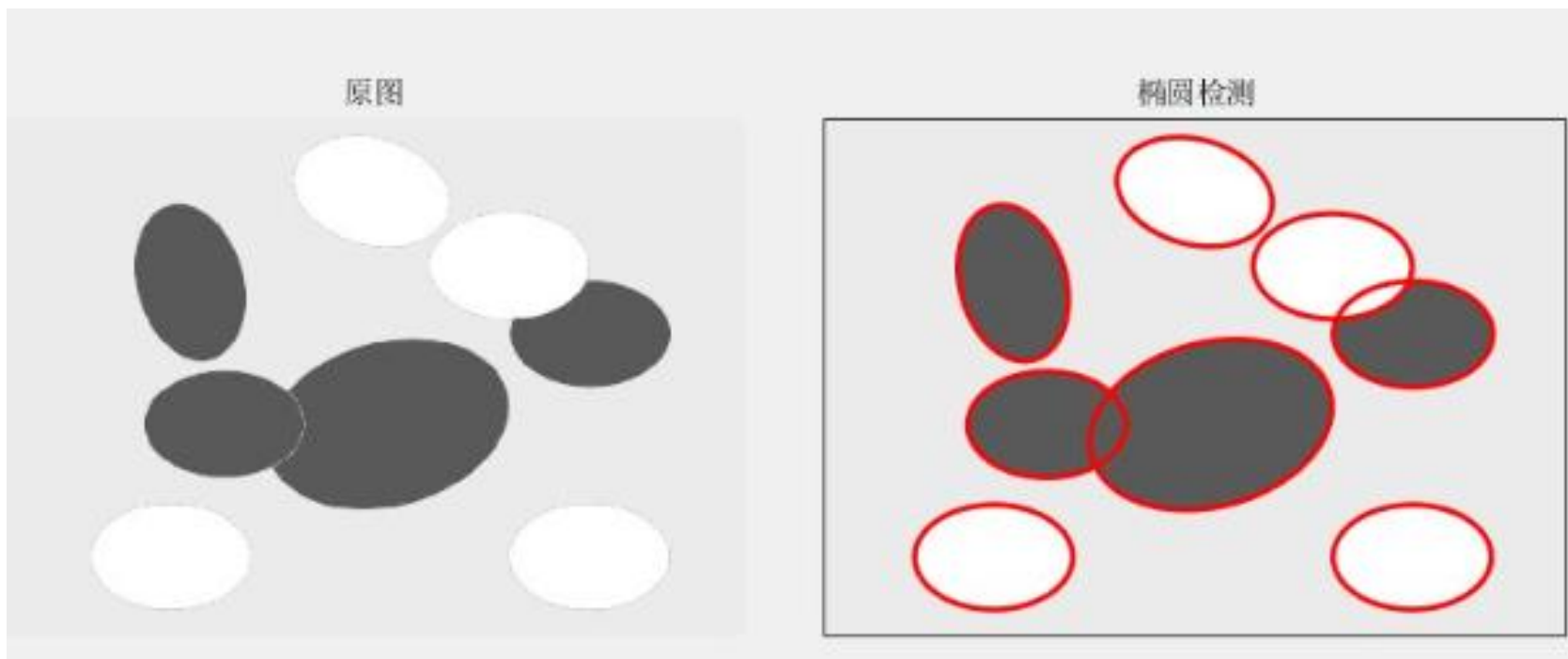


Transform parameter fitting

- RANSAC Line fitting example
- Repeat, until get a good result



基于随机抽样一致性的圆/椭圆检测

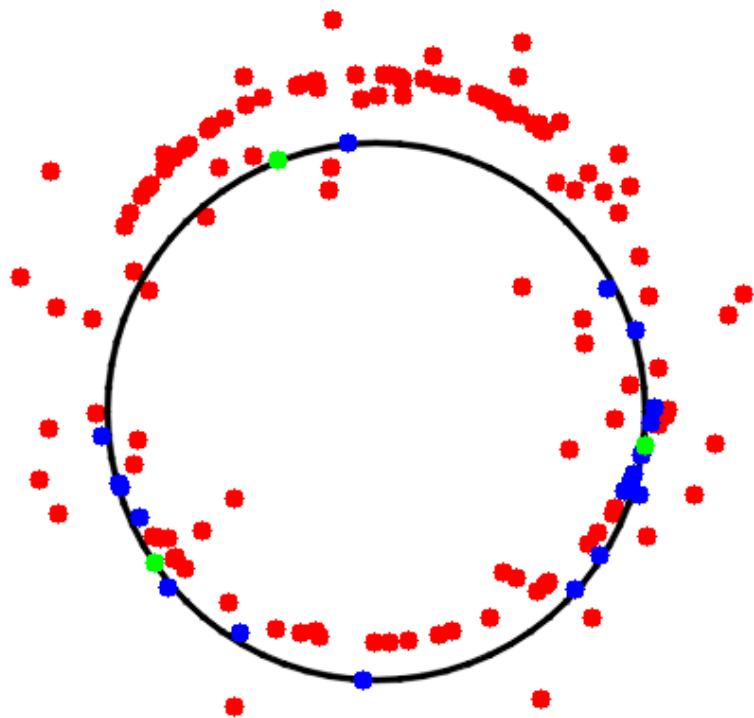


如果一个椭圆上所有的点都较严格地在该椭圆上（噪声较少），适合使用霍夫变换

基于随机抽样一致性的圆/椭圆检测



当噪声较多时，难以通过霍夫变换得到可靠的结果



可采用随机抽样
一致性进行参数
估计

随机选择一组点

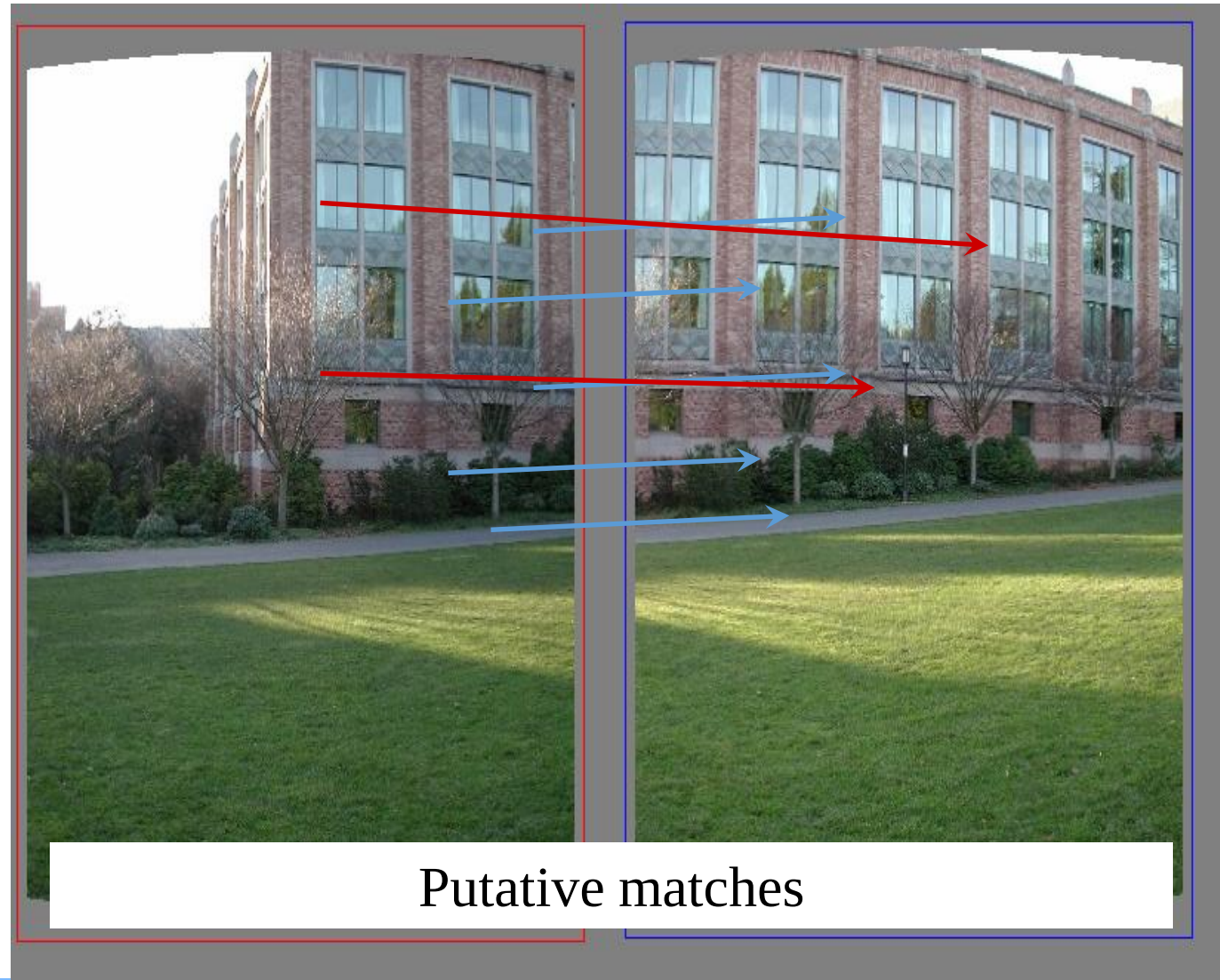
拟合椭圆/圆的参数

寻找与该椭圆/
圆相符的点

重新拟合参数

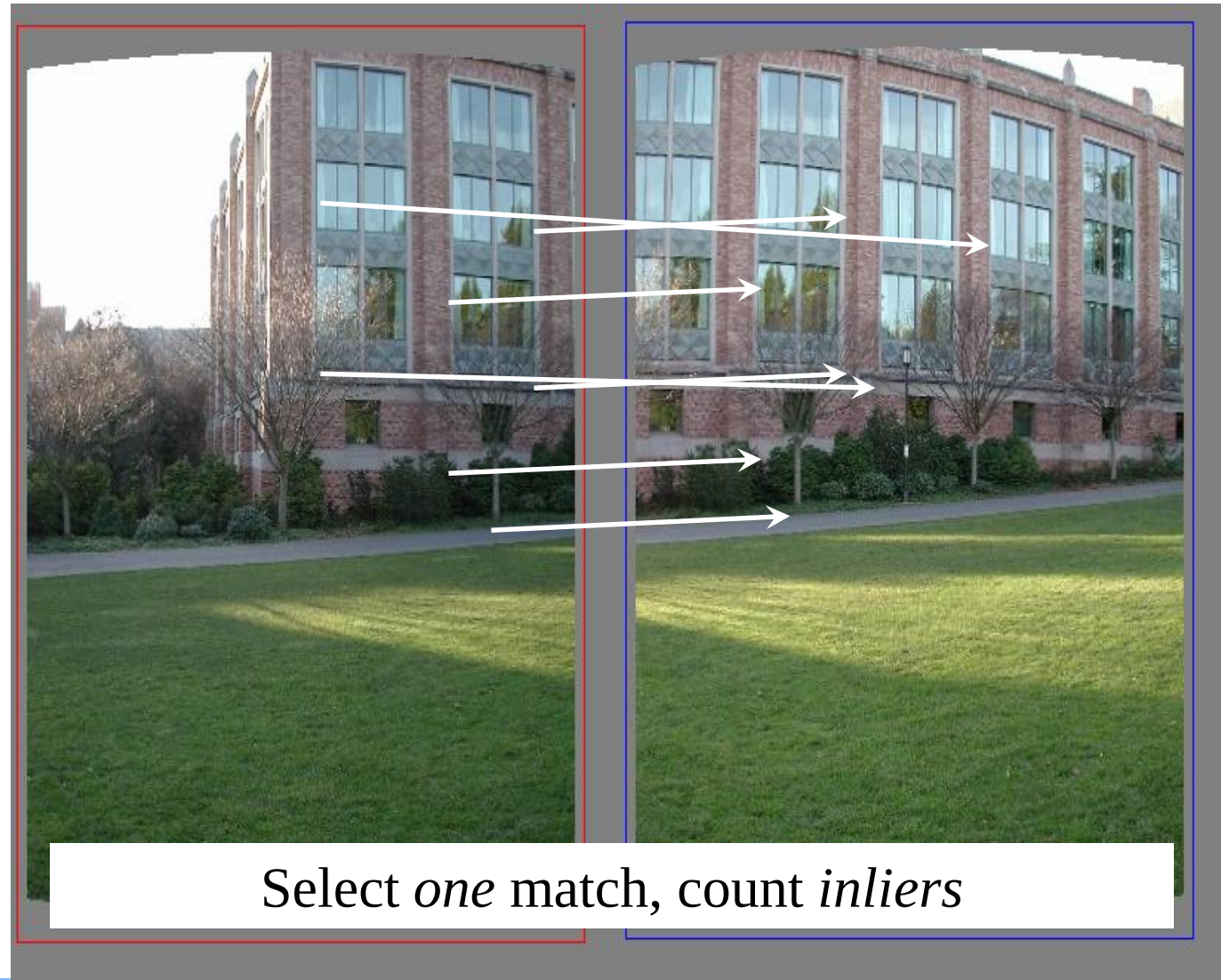
Transform parameter fitting

- RANSAC example



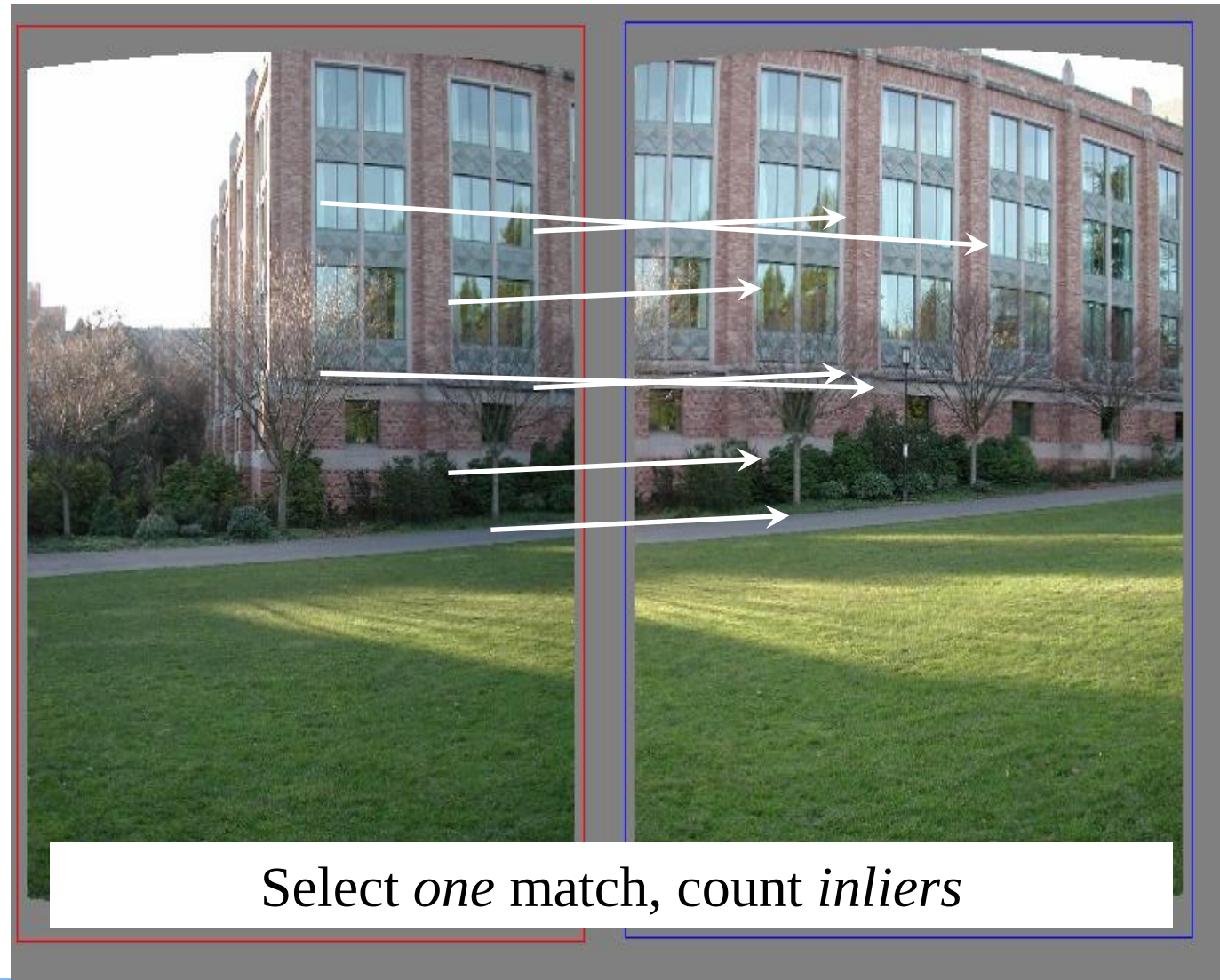
Transform parameter fitting

- RANSAC example



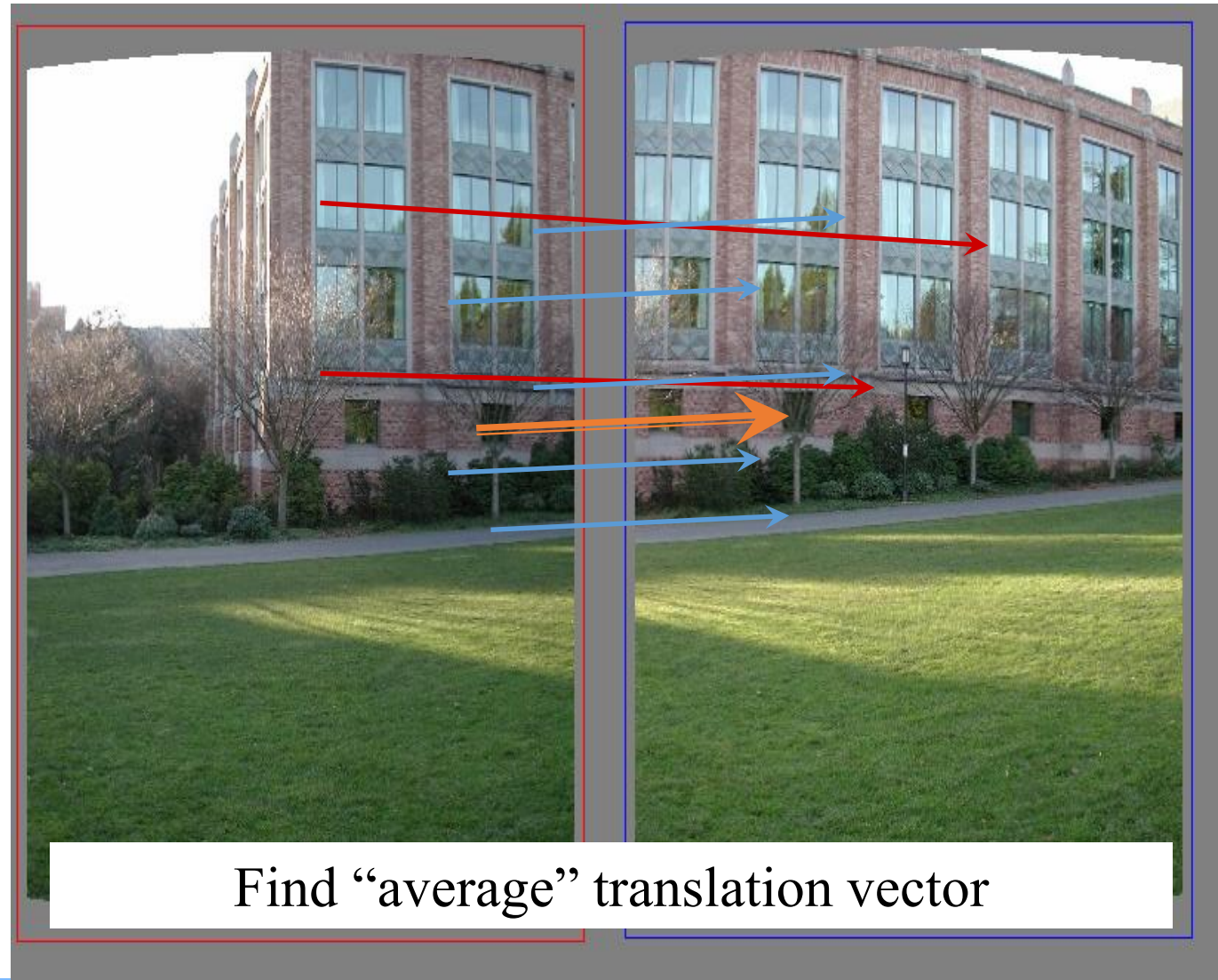
Transform parameter fitting

- RANSAC example



Transform parameter fitting

- RANSAC example

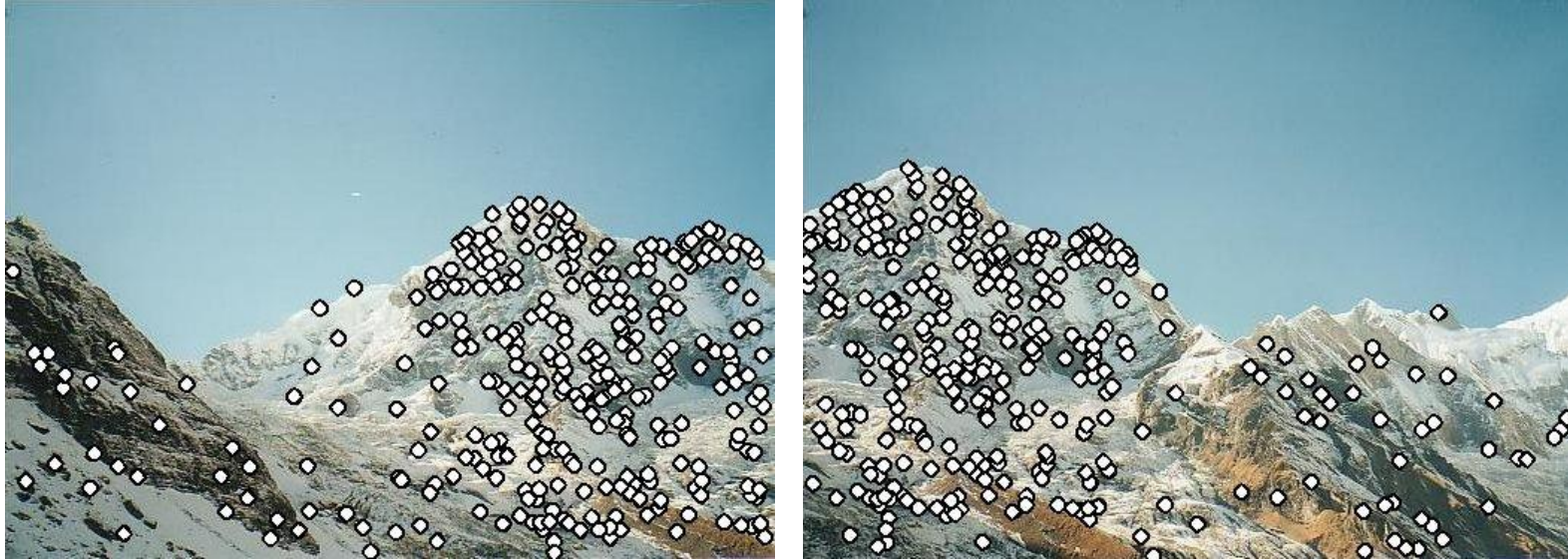


Feature-based alignment outline



- Two images

Feature-based alignment outline



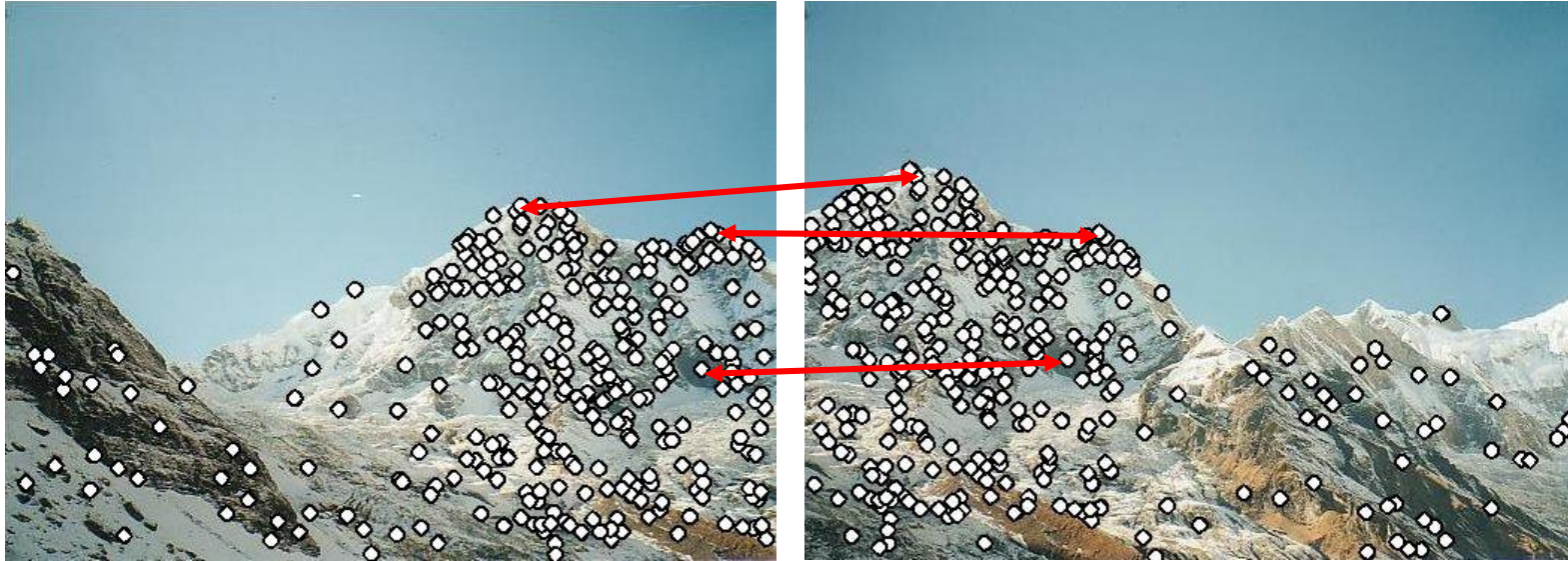
- Extract features

Feature-based alignment outline



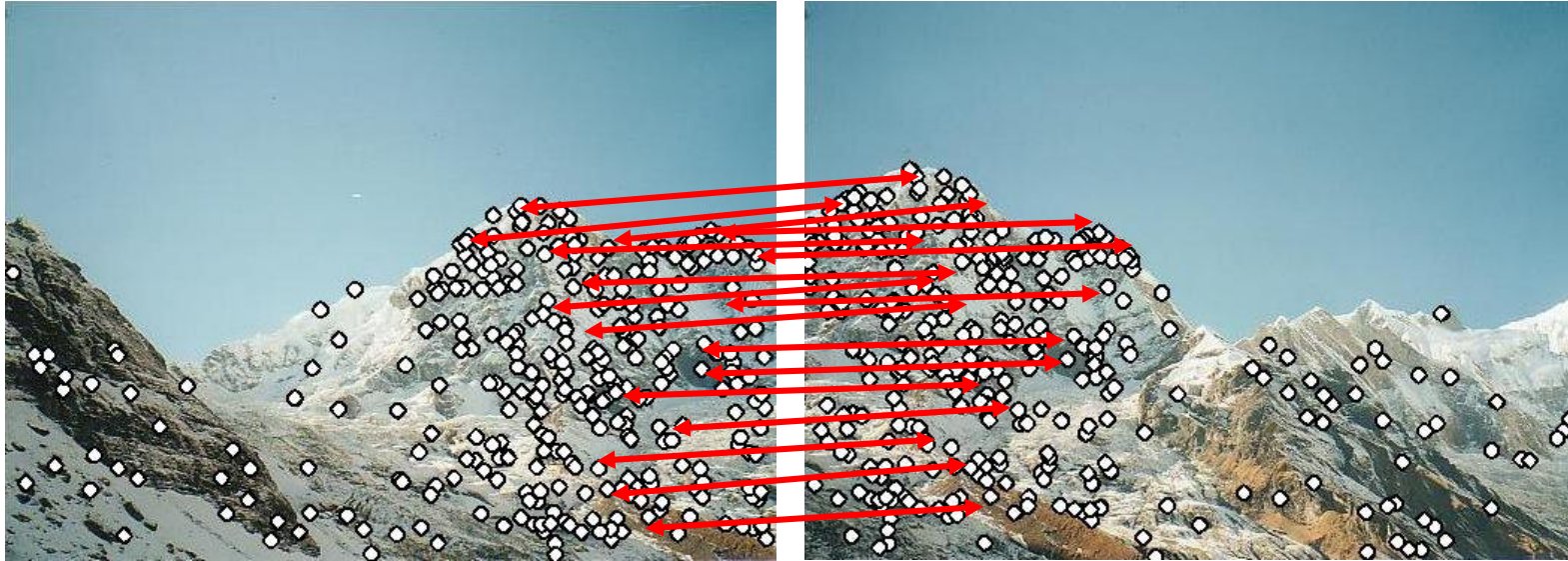
- Extract features
- Compute *putative matches*

Feature-based alignment outline



- Extract features
- Compute *putative matches*
- *Loop:*
 - *Hypothesize transformation T (small group of putative matches that are related by T)*

Feature-based alignment outline



- Extract features
- Compute *putative matches*
- *Loop:*
 - *Hypothesize transformation T (small group of putative matches that are related by T)*
 - *Verify transformation (search for other matches consistent with T)*

Feature-based alignment outline



- Transform the image and fusion